



AquaGuard AI

AI-Powered Water Safety Monitoring Platform

Capstone Project Report

SDG 6 — Clean Water and Sanitation

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United Nations Sustainable Development Goal 6

1. Selected SDG and Reason for Selection

SDG 6 — Clean Water and Sanitation

What SDG 6 Focuses On:

United Nations Sustainable Development Goal 6 aims to **"Ensure availability and sustainable management of water and sanitation for all"** by 2030. It addresses the global challenge of providing clean, accessible water and proper sanitation to every person. Key targets include:

- **Target 6.1:** Achieve universal and equitable access to safe and affordable drinking water.
- **Target 6.2:** Achieve access to adequate and equitable sanitation and hygiene for all.
- **Target 6.3:** Improve water quality by reducing pollution and eliminating dumping of hazardous materials.
- **Target 6.4:** Substantially increase water-use efficiency and ensure sustainable freshwater withdrawals.

Why This Issue Is Important:

Water is the most fundamental human need. Despite being essential for survival, **2.2 billion people worldwide still lack access to safely managed drinking water** (WHO/UNICEF 2023). Contaminated water is a leading cause of preventable diseases and deaths globally. Each year, approximately **485,000 people die from diarrheal diseases** caused by unsafe drinking water, with children under 5 being the most vulnerable.

In India, the Central Ground Water Board reports groundwater in several states is contaminated with fluoride, arsenic, iron, and nitrate beyond permissible limits. Urban areas face challenges from industrial effluents and aging pipelines. The lack of accessible, real-time water quality monitoring means many citizens consume contaminated water unknowingly.

This makes SDG 6 not just an environmental goal but a **public health imperative** — and one where AI can make a measurable difference.

2. Problem Statement

What Is the Problem?

Despite investments in water infrastructure, **citizens frequently encounter water quality issues** — discolored water, strange odors, unusual taste — without a quick, accessible way to:

1. **Assess the safety** of their drinking water in real time.
2. **Report water quality issues** to local authorities efficiently.
3. **Receive expert guidance** on contamination risks and preventive measures.

Currently, getting water tested requires physically collecting samples and sending them to a laboratory, taking days or weeks. During this time, affected families continue consuming potentially unsafe water.

Where Does This Problem Exist?

- **Urban areas in developing countries** (India, Bangladesh, Nigeria) where aging infrastructure causes contamination.
- **Rural and semi-urban communities** where water testing facilities are non-existent or prohibitively far.
- **Post-disaster and flood-affected regions** where water sources become contaminated.
- **Industrial zones** where factory effluents pollute groundwater.

Who Is Affected?

- **Households and families** dependent on municipal or borewell water.
- **Children and elderly** who are immunologically vulnerable.
- **Rural populations** without access to testing infrastructure.
- **Local government bodies** lacking efficient digital complaint systems.

Why Is It Serious?

- **Health impact:** Contaminated water causes cholera, typhoid, lead poisoning, and hepatitis A.
- **Economic burden:** Waterborne diseases cost developing nations billions annually.
- **Mortality:** Nearly 1.4 million people die each year from unsafe water (The Lancet, 2023).
- **Information gap:** Most people cannot identify unsafe water without expert analysis.

What Could Happen If Not Solved?

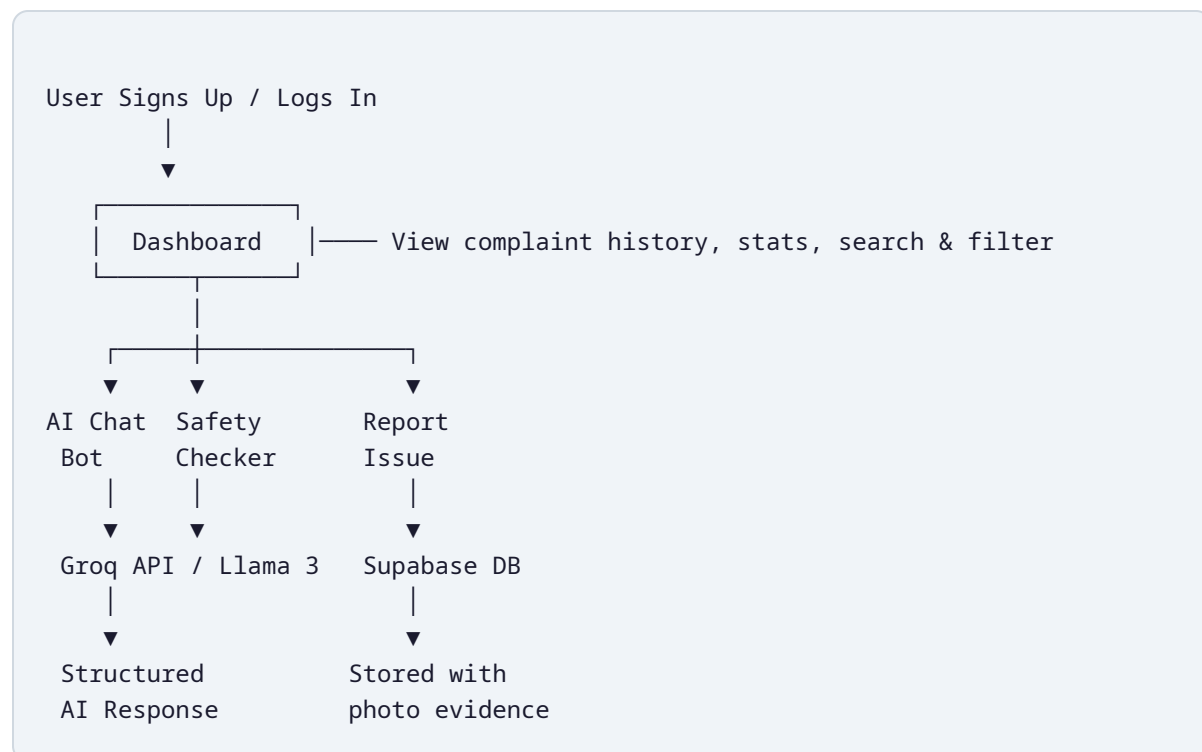
- Continued preventable deaths, particularly among children under 5.
- Widespread chronic health conditions from heavy metal exposure.
- Environmental degradation from unmonitored pollution.
- Failure to meet SDG 6 targets by 2030.
- Erosion of public trust in municipal water systems.

3. Proposed Solution

What Is AquaGuard AI?

AquaGuard AI is an AI-powered web application that empowers citizens to monitor, assess, and report water safety concerns in real time. It leverages **Generative AI (Llama 3 via Groq API)** to provide instant expert-level analysis and recommendations, eliminating the need for expensive laboratory tests for initial assessment.

How the System Works



How It Helps Solve the Problem

Problem	AquaGuard AI Solution
No quick way to assess water safety	AI-powered instant analysis via chatbot and safety checker
Reporting issues is cumbersome	Digital complaint form with photo upload and tracking
Lack of expert guidance	AI provides structured expert-level advice 24/7
No tracking of complaints	Dashboard with real-time status tracking and search
Information gap for citizens	AI education on water safety and prevention

Who Will Use the Application

- **Citizens and households** checking water safety before consumption.
- **Community health workers** screening water quality in their areas.
- **Local municipal bodies** receiving and tracking citizen complaints digitally.
- **Students and researchers** studying water quality and environmental health.

4. Project Features

4.1 User Authentication System

Secure authentication with **Login and Registration** functionality. Users create accounts with email and password, with session management by Supabase Auth. Includes password visibility toggle, remember-me, and automatic session persistence. Protected routes ensure only authenticated users access internal pages.

4.2 User Dashboard

The **central hub** displaying a personalized welcome, summary statistics (Total Complaints, Pending, Resolved) in visual cards, and a paginated complaints table. Users can **search** by issue type or location and **filter** by status. Updates in real time as new complaints are submitted.

4.3 AI-Powered Chatbot (AI Assistant)

A conversational interface powered by **Llama 3.1-8B** (via Groq API) acting as a water safety expert. Answers questions about contamination, purification, waterborne diseases, and conservation. Responses are structured into **Possible Cause, Safety Advice, and Preventive Measures**. Chat history is persisted and displayed in a sidebar.

4.4 Water Safety Checker

An **interactive diagnostic tool** where users describe their water's color, smell, and taste through intuitive selection buttons. The AI analyzes the combination and returns a **risk assessment** (Low/Medium/High) with the likely cause and recommended action, displayed with color-coded risk badges.

4.5 Complaint Reporting System

A **structured complaint form** with fields for Issue Type, Location, Description, and **Photo Evidence upload** (PNG, JPG, GIF up to 10MB). Images stored in Supabase Storage. Upon submission, users receive a unique reference ID and confirmation.

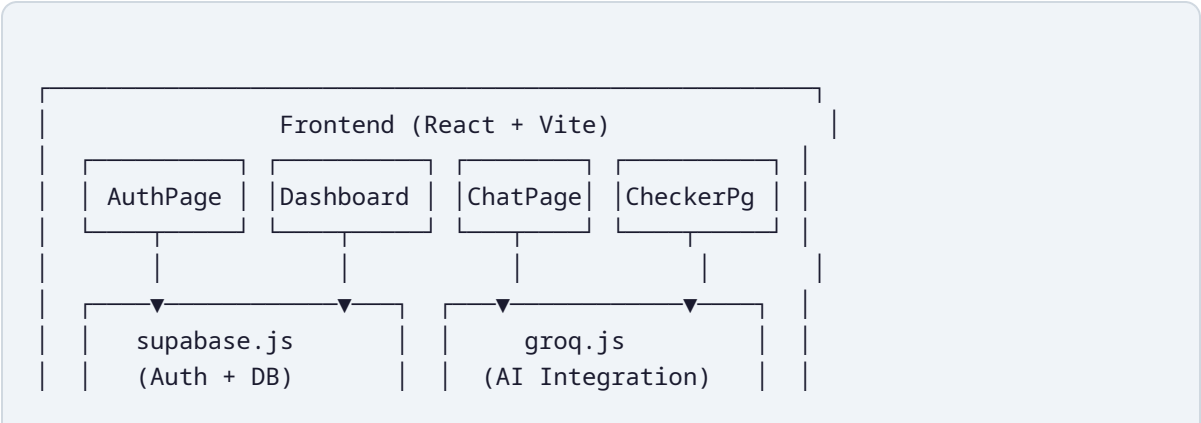
4.6 Responsive Navigation and Layout

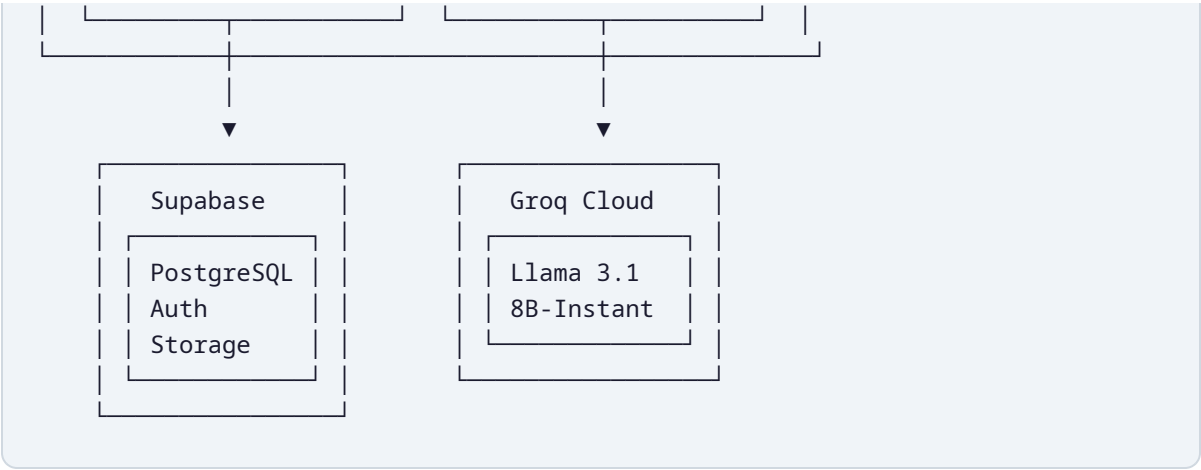
A **Material Design-inspired navigation bar** with active route highlighting, user account icon, and logout. Footer includes links to Privacy Policy, Terms of Service, and SDG 6 info. Fully responsive across desktop, tablet, and mobile.

5. Technology Stack

Layer	Technology	Purpose
Frontend Framework	React 19	Component-based UI development
Build Tool	Vite 8	Fast dev server and optimized builds
Routing	React Router DOM v7	Client-side routing with protected routes
Styling	Tailwind CSS v3.4	Utility-first CSS with custom design tokens
Generative AI / NLP	Groq API + Llama 3.1-8B	AI-powered water safety analysis and chatbot
Backend / Database	Supabase (PostgreSQL)	Authentication, data storage, real-time queries
File Storage	Supabase Storage	Photo evidence uploads for complaints
Authentication	Supabase Auth	Email/password auth with session management
Icons	Google Material Symbols	Modern icon system throughout UI
Deployment	Vite Build	Production-ready static site

Architecture Diagram





6. Project Screenshots (Proof of Work)

6.1 Login Page

Authentication page with animated wave decorations, AquaGuard AI branding, and Login/Register toggle.

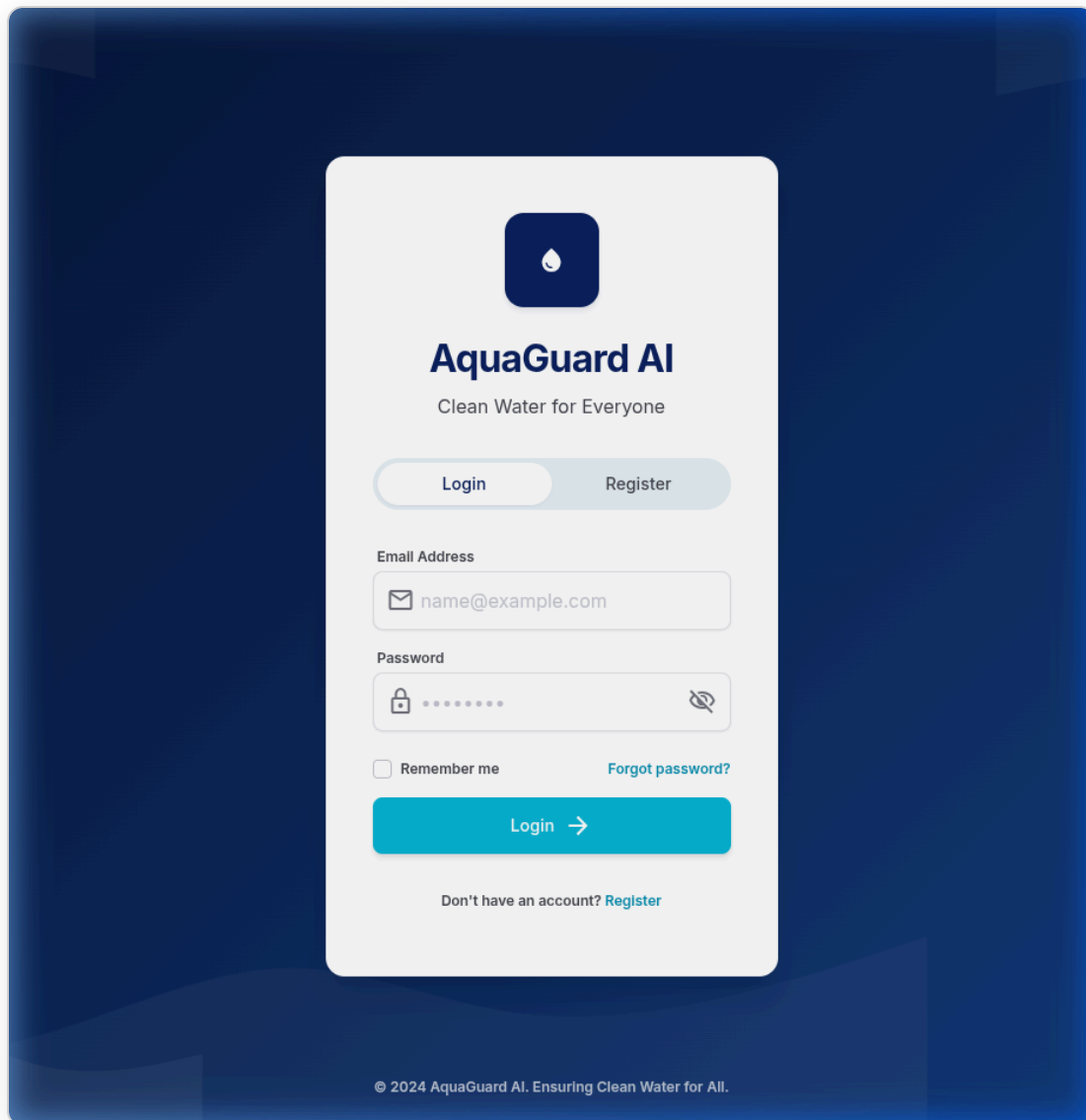


Figure 1: AquaGuard AI — Login Page

6.2 Registration Page

Registration view for new user account creation with email and password.

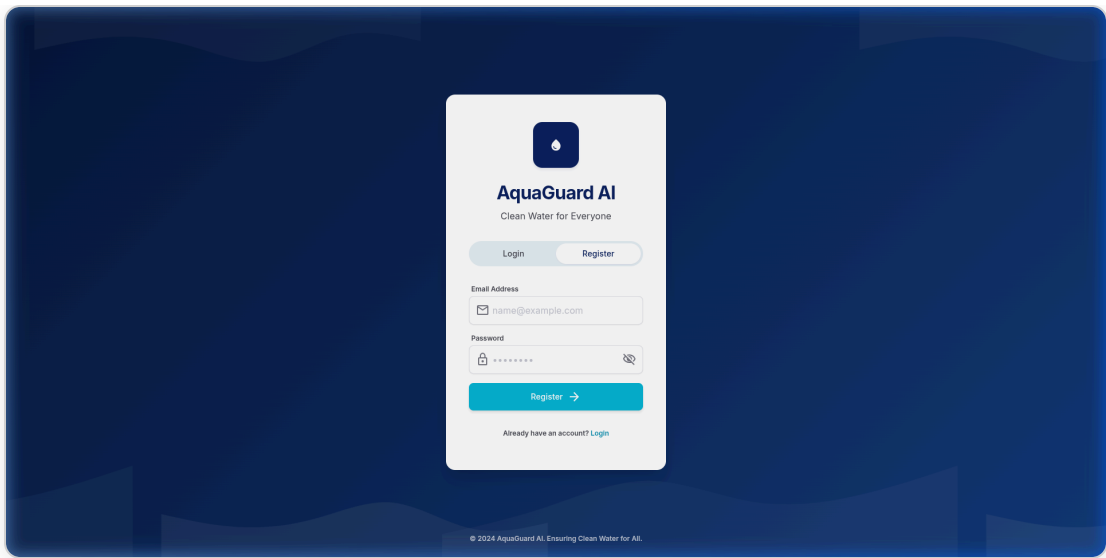


Figure 2: AquaGuard AI — Registration Page

6.3 User Dashboard

Main dashboard showing welcome message, complaint statistics, and complaints table with search and filter.

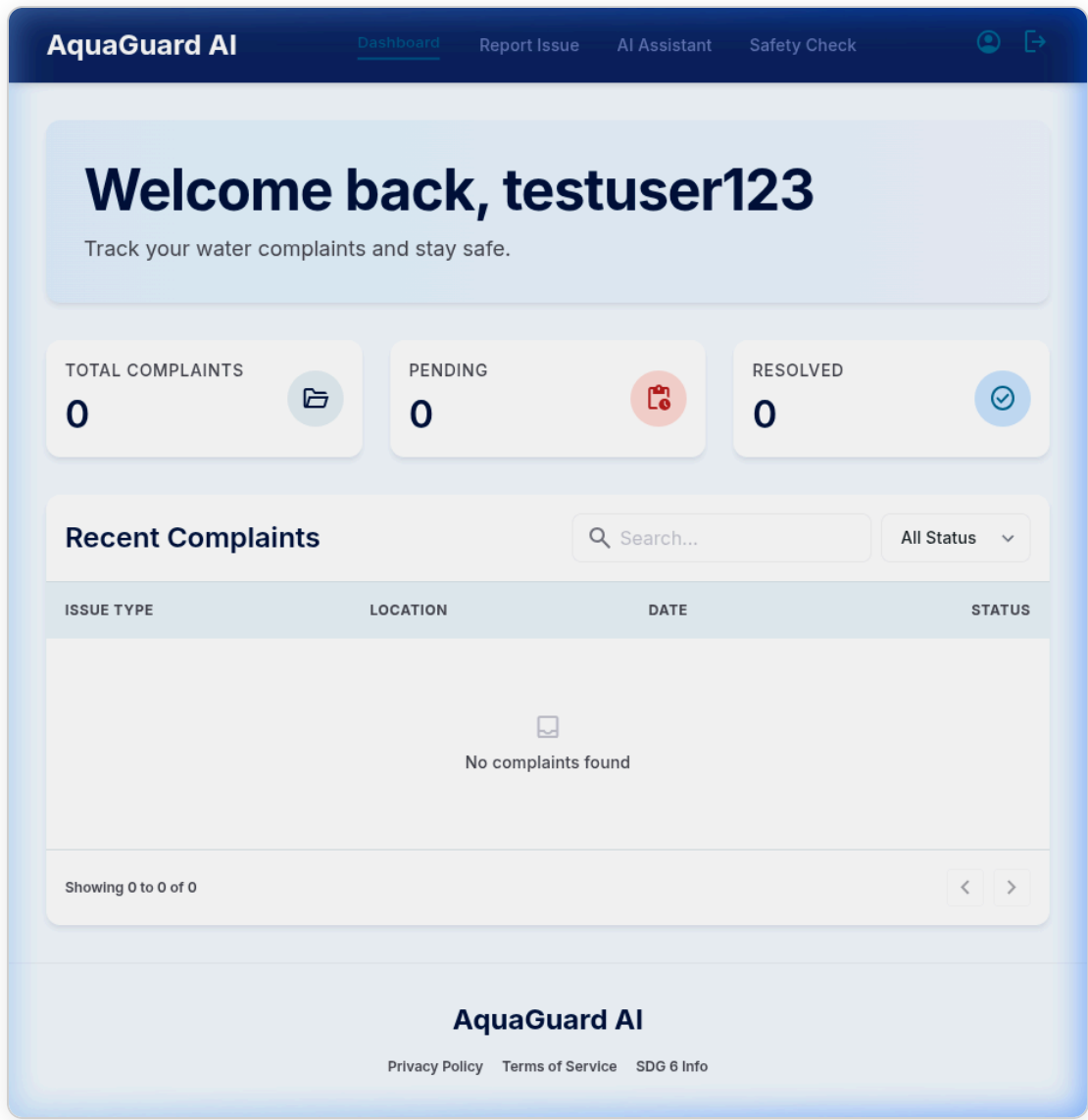


Figure 3: AquaGuard AI — User Dashboard

6.4 Report a Water Issue

Complaint submission form with issue type selector, location input, description, and photo upload.

AquaGuard AI

Dashboard

Report Issue

AI Assistant

Safety Check

Report a Water Issue

Please provide details about the water safety concern to help us investigate quickly.

Issue Type

Select the type of issue

Location

Enter street address or coordinates

Description

Describe the issue in detail...

Photo Evidence

Click to upload or drag and drop

PNG, JPG, GIF up to 10MB

Submit Complaint

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Figure 4: AquaGuard AI — Report Issue Form

6.5 AI Assistant — Chat Interface

AI chatbot with chat history sidebar, greeting message, and input area. Powered by Llama 3.

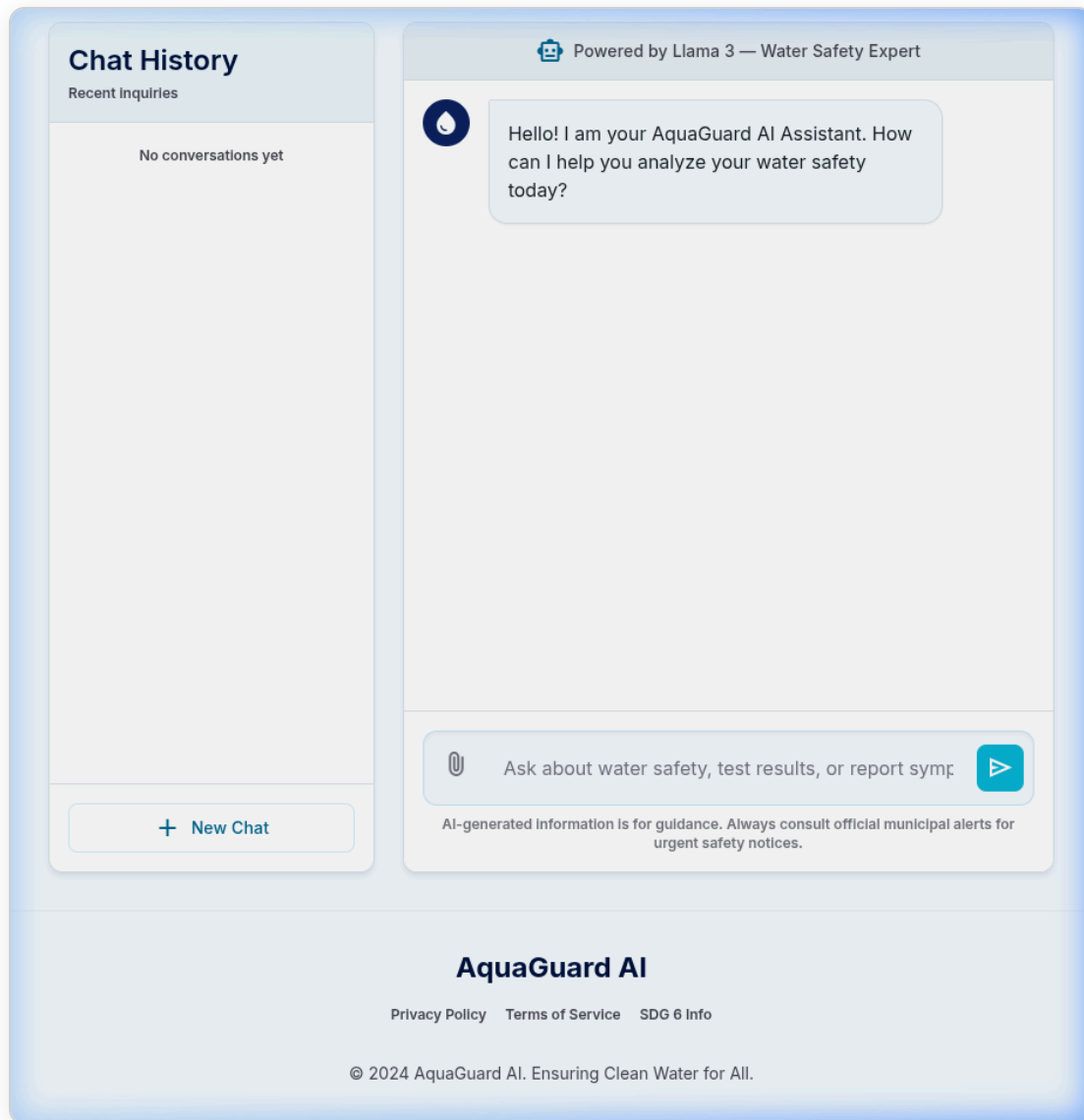


Figure 5: AquaGuard AI — AI Assistant Interface

6.6 AI Assistant — Live Response

Real interaction: user asks about yellow, metallic-smelling water. AI responds with structured sections.

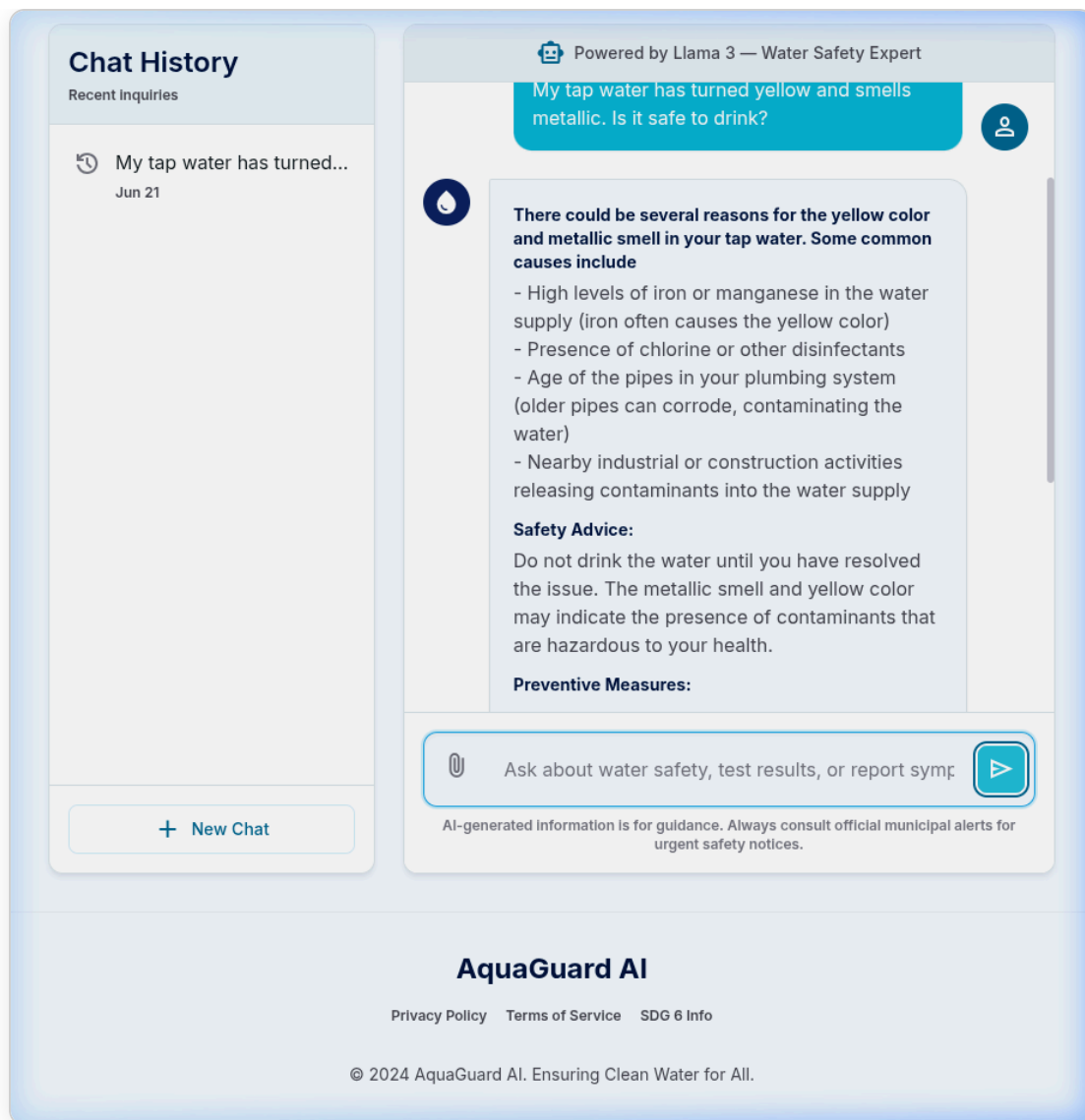


Figure 6: AquaGuard AI — Live AI Response with Safety Advice

6.7 Water Safety Checker — Input

Interactive water analysis tool with color swatches, smell options, and taste options.

AquaGuard AI

DashboardReport IssueAI AssistantSafety Check

Water Safety Checker

Describe your water to get instant AI analysis and actionable safety recommendations aligned with SDG 6.

Water Color

Select the closest match to what you see.

Clear

Yellow

Brown

Greenish

Cloudy/White

Smell

Identify any distinct odors.

None

Chlorine

Earthy / Musty

Rotten Egg

Metallic

Chemical

Taste

If safe to do so, describe the taste.

Normal / None

Salty

Metallic

Bitter

Sweet

Analyze Water Safety

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Figure 7: AquaGuard AI — Water Safety Checker Input

6.8 Water Safety Checker — AI Results

After selecting Yellow/Metallic/Metallic, AI returns **HIGH RISK** with cause and recommendation.

Select the closest match to what you see.

Clear

Yellow

Brown

Greenish

Cloudy/White

Identify any distinct odors.

None

Chlorine

Earthy / Musty

Rotten Egg

Metallic

Chemical

If safe to do so, describe the taste.

Normal / None

Salty

Metallic

Bitter

Sweet

🔬 Analyze Water Safety

⊗

HIGH RISK

Do Not Consume

Check Again

🔧 Likely Cause

Presence of high levels of iron or copper in the water

🛡️ Recommendation

Have your water tested for heavy metal contamination and consider installing a home filtration system

Report to Local Authority

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Figure 8: AquaGuard AI — HIGH RISK Analysis Result

7. Open Source Repository

GitHub Repository: *(Insert your GitHub repository URL here)*

Example: `https://github.com/yourusername/aquaguard-ai`

The repository contains:

-  Complete source code (React + Vite)
-  Project documentation (README.md)
-  Environment configuration template (.env.example)
-  Package dependencies (package.json)
-  Build configuration (vite.config.js, tailwind.config.js)
-  .gitignore for proper version control

Important: Before submission, push your code to GitHub and verify the repository link is accessible (set to Public).

8. Future Scope

8.1 Mobile Application

Develop a **React Native or Flutter mobile app** for Android and iOS, enabling users to report issues and check water safety from their phones.

8.2 IoT Sensor Integration

Integrate with **IoT water quality sensors** (pH, TDS, turbidity) for automated, real-time monitoring feeding directly into the AI analysis engine.

8.3 Geospatial Mapping and Heatmaps

Implement a **geographic heatmap** visualizing water quality complaints to help authorities identify contamination hotspots.

8.4 Multi-Language Support

Add **regional languages** (Hindi, Marathi, Bengali, Tamil) to serve non-English-speaking populations.

8.5 Advanced AI Features

- **Image analysis** using computer vision to detect contamination from uploaded photos.
- **Predictive analytics** using historical data to predict contamination events.
- **Voice-based interaction** for users with limited literacy.

8.6 Government and NGO Integration

Build APIs for **municipal water authorities and NGOs** to manage citizen complaints, creating a citizen-to-government feedback loop.

8.7 Offline Support (PWA)

Implement **Progressive Web App** capabilities with offline caching for areas with poor connectivity.

File Name: SDG_AquaGuardAI_AdityaDhembare.pdf

Submission Portal: <https://leapnextgenscholar.in/>